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ORIGINAL PAPER

Temperature- and solvent-dependent migrations of di(2-ethylhexyl) phthalate, the hazardous plasticizer from commercial PVC blood storage bag

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Abstract Towards a process development for the bioremediation of phthalic acid esters (PAE) blended in various plastics, we explored the type of PAE in commercially available PVC blood storage bags (BB), its solvent- and temperature-dependent migrations and coefficient of diffusion (D) behaviors at static condition by solvent extraction method. The PAE [about 34 % (*w*/*w*)] in BB was confirmed as DEHP by UV, TLC and GC-MS analyses. When effects of five solvents on the rate of migration of intact DEHP from BB tested (40 °C), order of migration during the initial hrs was: *n*-hexane>methanol>*t*-butyl alcohol>isopropyl alcohol>ethanol. At 40 °C, about 60 % (of total) DEHP in BB migrated into *n*-hexane, which rose to 80 % at 60 °C and almost 95 % at boiling (69 °C) in the first h. DEHP diffusion from BB was dependent on its concentration and nature of surrounding medium and temperature. Fickian approximation was used to calculate D values for DEHP migration, and in *n*-hexane average values were: 1.26×10^{-7}, 1.92×10^{-7} and 2.18×10^{-7} cm^2 sec^{-1} at 40, 50 and 60 °C, respectively. This study shows that *n*-hexane is the preferred solvent to extract DEHP from used or waste medical and allied PVC products for small scale, especially screening studies. Briefly, DEHP and solvent were separated individually upon distillation of the extract; thus DEHP, microporous PVC BB sheet and solvent were recovered for reuse inexpensively without environmental contamination.

Keywords PVC blood bag · DEHP migration · TLC · GC-MS · SEM · AFM

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Introduction

Di(2-ethylhexyl)phthalate (DEHP) is the predominant phthalic acid ester (PAE) mainly used to plasticize polyvinyl chloride (PVC) thermoplastics [1–3]. Since DEHP is bound only physically on to the PVC polymer or embedded in the mesh spaces of its net, it can easily be leached into the surrounding medium, and after cycling through the ecosystem it ultimately reaches in to humans [4, 5]. DEHP very poorly soluble in water, but dissolves efficiently in oil or fatty foods, and thus its highest concentrations have been detected in milk, cheese and fat deposits [6].

Though the usage of DEHP has partially or completely been banned in several countries, especially in toys, still it is the leading plasticizer being used in PVC-borne medical devices [7, 8]. The PVC used in medical devices like intra-venal fluid bags including blood storage bags (BB), tubing, syringes and catheters are often composed of 30 to 40 % (wt) DEHP [9]. DEHP thus reached into human system gets accumulated in tissues, blood cells and body fluids [10, 11], which in turn shown to cause hepatomegaly, reproductive

and developmental toxicity and are suspected to be endocrine disruptors and potent carcinogens [10–13].

Concerns have therefore been raised by environmental groups that DEHP exposure may result in gender birth defects and the feminisation of boys [14]. DEHP, in the form of its monoethyl metabolic intermediate, acts through a receptor-mediated signaling pathway to suppress estradiol production in the ovary, leading to anovulation [15]. Migration of DEHP from the DEHP-PVC BB into the stored blood and blood components has much been studied [16, 17]. Amidst these health concerns, owing to their flexibility, convenience, cost effectiveness and handy nature, usage of DEHP-plasticized PVC in the medical devices has become rampant.

Factors like temperature, pressure, pH, presence of solvents, radiations, organic compounds like drugs, etc., are the supposed factors which may accelerate DEHP migration in to the surrounding environments. Kim et al. [4] observed that the release behavior of DEHP from flexible PVC sheet is affected by the nature of surrounding medium and temperature, and the diffusion coefficient at high temperature is higher than that at low temperature. Supercritical fluid extraction using CO_2 [18] or microwave-assisted extraction [19] has been reported as efficient extraction techniques for the extraction of phthalates from PVC on large scale. Alternatively, Haishima et al. [20] demonstrated a spectrophotometric assay using methyl yellow as an indicator for the release of DEHP from the medical devices, i.e, lipophilic medicines releasing large amount of DEHP showed high solubility of methyl yellow, while hydrophilic medicines showed low solubility. Though efficient over conventional solvent extraction method, for screening of compounds at small scale and simple analyses, conventional solvent extraction system would be advantageous in terms of cost and easiness.

No study is seen in literature showing the effect of temperature and solvents in the migration of DEHP from the commercial BB. In the light of the aforesaid background, using the conventional extraction system, a pioneering study was accomplished with the focus: (1) to confirm the type and quantity of PAE present in the leading brands of commercially available BB in Kerala, India; (2) to understand the behavior of PAE migration from PAE-plasticized commercially available PVC BBs into various solvents under varying temperature thereby choosing the preferred solvent for extraction of PAE from the used or waste PVC medical devices and allied products; (3) to confirm physically that PAE is removed from the PVC BB by analyzing the surface patterns employing SEM and AFM techniques; and (4) to concentrate the extracted PAE from the solvent used for extraction, thus to make PAE, solvent and PVC as separate components without contaminate the environment.

Materials and methods

Materials

All chemicals and reagents used in this study were of analytical reagent grade, and procured from Merck India P. Ltd. All glass wares used in this study were procured from Borosil, India. The commercial PVC blood storage bags (specimens) were obtained from three leading firms, viz., HL haemopack (HPB) from Hindustan Latex Ltd, Thiruvananthapuram, India (Batch No. HO 30419B, June 2009), Medibag (MBB) from Eastern Medikit Ltd, Mumbai, India (Batch No.Ch-B 81004A, January 2008) and Terumopenpol (TPB), Thiruvananthapuram, Kerala, India (Batch No. 191082S, September 2008).

Extraction and quantification of DEHP contained in the BBs

The anti-coagulant solution contained in the whole BB was drained out and washed thoroughly with double distilled water. PVC BB specimens were cut into small pieces of square size (7 mm^2), and 25 mg BB pieces were transferred into a refluxing apparatus containing 25 mL of *n*-hexane and refluxed for 3 h at 69 °C. The quantity of DEHP extracted into *n*-hexane from each BB specimen was quantified using DEHP standard graph.

Migration study

For standard preparation, the optical density (OD) of known concentrations of DEHP in *n*-hexane, methanol, ethanol, isopropyl alcohol or *t*-butyl alcohol were measured using a UV-Vis Biospectrophotometer (Elico BL 200, India) at 275 nm with inbuilt peltier system. The migration behavior of DEHP from BB into *n*-hexane was studied at 40, 50 and 60 °C; while that in methanol, ethanol, isopropyl alcohol and *t*-butyl alcohol was carried out at 40 °C. 1 mg BB per mL each of these solvents was used for quantifying the migrated DEHP (in standard flask) at definite time intervals in a thermo-stated water bath. Migration of DEHP was analyzed (OD at 275 nm) continuously at definite time intervals. Samples were withdrawn at every 5 min during the first 1 h and then in every 1 h interval. Immediately after OD measurement, the sample poured back in the migration system so that the total volume was kept constant. The weight of migrated DEHP in solvents was calculated by fitting the values on to the standard graph.

Thin layer chromatography (TLC)

TLC grade Silica Gel-G was used for the rapid screening of DEHP in BB extract. A solvent mixture containing *n*-hexane and ethyl acetate in the ratio 9:1, respectively was used

for the development of the sample. It was visualized by keeping the chromatogram in an iodine chamber for 30 min.

Gas chromatography-mass spectrometry (GC-MS)

To analyze the presence of migrated DEHP in the solvent, AGILENT 6890N GC with 5975 inert MSD was used. The column used was HP5-MS (30 m×0.25 mm×0.25 μm). The carrier gas was helium at a flow rate of 0.7 mL/min. The m/z scan range was 40–450 m/z. Initial column temperature: 80 °C (hold 4 min), increased to 300 °C (hold 4 min) at 10 °C/min. Injection temperature: 250 °C. Sample analysis was done at Interfield Laboratories, Kochi, Kerala, India.

Scanning electron microscopy (SEM)

The surface patterns of the PVC BB pieces before and after DEHP extractions were analyzed using the SEM (HITACHI, SU 6600, Japan) facility available at the National Institute of Technology (NIT), Calicut, India.

Atomic force microscopy (AFM)

The topographic patterns of the PVC BB pieces before and after DEHP extractions were analyzed employing AFM (Park Systems, XE-100, Korea) facility available at the NIT, Calicut, India, and measurements were performed by a noncontact mode so that the cantilever vibrated near the surface without physical contact between the tip and the sample.

Concentration of PAE

Known quantity of BB was put in boiling solvent of measured volume for 1 h. The extract was then distilled to separate PAE from solvent, thus to make them independent components.

Statistics

The Fick's law was used to calculate the diffusion coefficient, employing the formula, $M_t/M_\infty = 2\left(\sqrt{Dt/\pi l^2}\right)$, where M_t represents the total amount of DEHP released at time t and M_∞ the amount of DEHP originally present in the specimen (BB), D is the diffusion coefficient and l denotes the thickness of the specimen. Phthalate migration (%) bound to BB was calculated using the formula, $\% = \frac{C_t \times 100}{C_i}$, where C_i is the total phthalate contained in the BB and C_t is the quantity of phthalate migrated in to the surrounding solvent in time, t. The data presented here are average values of at least three independent values with±S. D. Microsoft Excel was used to plot the graph.

Results

Quantification and screening for DEHP in BB

Our first objective was to prove whether the PAE present in the BB was DEHP. After extraction, the total DEHP present in all the three PVC BBs, viz., HPB, MBB and TPB, used in this study was quantified (by fitting in to the standard graph of DEHP), and it was observed HPB, MBB contained only DEHP as the plasticizer, whose average was 34 % (*w*/*w*). i.e., HPB contained 35.48±0.30 %, while MBB contained 33.3±0.40 % DEHP by weight. Since TPB contained a PAE mixture including DEHP, which was excluded from further studies. The presence of DEHP in all the three PVC BBs was initially measured using UV spectrometry for the single characteristic peak at λ 275 (Fig. 1). As shown in Fig. 2, the extracted PAE from HPB and MBB showed a single clear spot on TLC chromatogram, with an R_f value 0.71.

DEHP in PVC BB confirmed by GC-MS

The presence of DEHP was further confirmed by GC-MS analyze. Figure 3a and b show the GC-MS profile of extracted DEHP in *n*-hexane from HPB. GC profile (Fig. 3a) shows a sharp peak at RT 24.54 min with a corresponding m/z, 149. The MS fragmentation spectrum (Fig. 3b) of the GC peak at RT 24.54 min revealed by molecular mass and standard DEHP analyses that the spectra was of DEHP.

Pattern of DEHP migration from BB

Figure 4a & b illustrate the comparison of the DEHP migration profiles of all two BB in *n*-hexane at 40 °C. In comparison, total DEHP content present in PVC BB was found proportional to the rate of migration. Faster migration

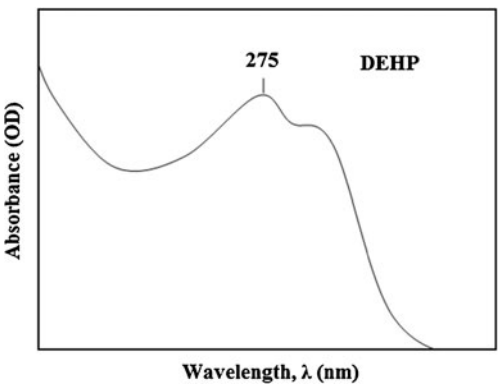


Fig. 1 UV profile of the extracted DEHP in *n*-hexane from the PVC-BB MBB. It shows a characteristic peak of DEHP at 275 nm



Fig. 2 TLC chromatogram shows single clear spot with an R_f value 0.71 for standard DEHP and extracted DEHP from two PVC-BB, HPB and MBB. BB. Samples were in *n*-hexane, and the *n*-hexane/ethyl acetate (9:1) solvent system was used for development. Iodine vapor chamber (30 min) was used for visualization

observed with the BB HPB, which contained 35.48±0.30 % DEHP by weight and bit slower migration with MBB, which contained 33.3±0.40 % DEHP by weight. Though the migration pattern was monitored up to 48 h, the rate of migration was leveled off at about 9th h (Fig. 4a). Almost 50 % of the total DEHP in the BB was migrated into the solvent at the initial 30 min (Fig. 4b).

Effect of solvent on migration

The cumulative amount of DEHP released varied with solvent nature (Fig. 5). We have analyzed the migration profile of the two brands of BB in four different alcohols at 40 °C. The result with MBB is shown in Fig. 5. Of them, *t*-butyl alcohol emerged as the good solvent for facilitating faster DEHP migration at 40 °C (~77 % released) and lesser influence was shown by ethanol (~52 %). Methanol and isopropyl alcohol showed comparable influences on DEHP migration at 40 °C (~61 %). Interestingly, as seen in Fig. 5, the initial rate of migration with *t*-butyl alcohol was comparably less than those of other three solvents. However, most of the DEHP in the BB migrated out in the initial h itself (Fig. 5).

Effect of temperature on DEHP migration

The effect of temperature (at 40 °C, 50 °C and 60 °C) on DEHP migration from two brands of BB was investigated. As in Fig. 6a, migration of DEHP from HPB was highest at 60 °C (~93 %), while it was least at 40 °C (~80 %). It is very clear that the cumulative amount of DEHP released at the first 30 min was linear in nature at all temperatures (Fig. 6b).

Diffusion coefficient, D

The Fick's law was used to calculate the diffusion coefficients (D) from the diffusion profile of DEHP into the solvents. It tells us the diffusion of DEHP into the solvent and counter diffusion of solvent back into the PVC, which in turn dependent on the temperature and the surrounding solvent nature. The release and diffusion of DEHP from the BB were based on the given assumption that the volume of surrounding solvent is very large compared to the amount of released DEHP. The migration of DEHP from all the sides of the BB was considered. Figure 7a clearly shows that DEHP migration (in *n*-hexane) depends on its concentration in the BB and time. And thus, the functional relation of Mt/M_∞ and $\sqrt{t}$ demonstrated good linearity at the initial stage of the migration profile (Fig. 7b), i.e., Mt/M_∞ was increased with time during the initial 1 h and then leveled off. This relationship is further evident from the Table 1.

Effect of solvents on D

The D value of DEHP in BB varies with the surrounding medium. Of four alcohols tested, methanol was emerged as the best alcohol to increasing the rate of migration (Fig. 8a & b). In fact, *t*-butyl alcohol showed less activity during the initial hrs and then elevated and overcome the effect of methanol. This difference is further clarified in Table 2.

SEM pattern of BB

Before extraction, the BB was soft and flexible. But after complete removal of DEHP by extracting with *n*-hexane, the BB became rough and stiff. Figure 9 shows the SEM profile of the outer surface of BB (HPB) before (A) and after (B) DEHP extractions. Before extraction, the surface and PVC mesh spaces are seen tightly packed with DEHP and even DEHP aggregates

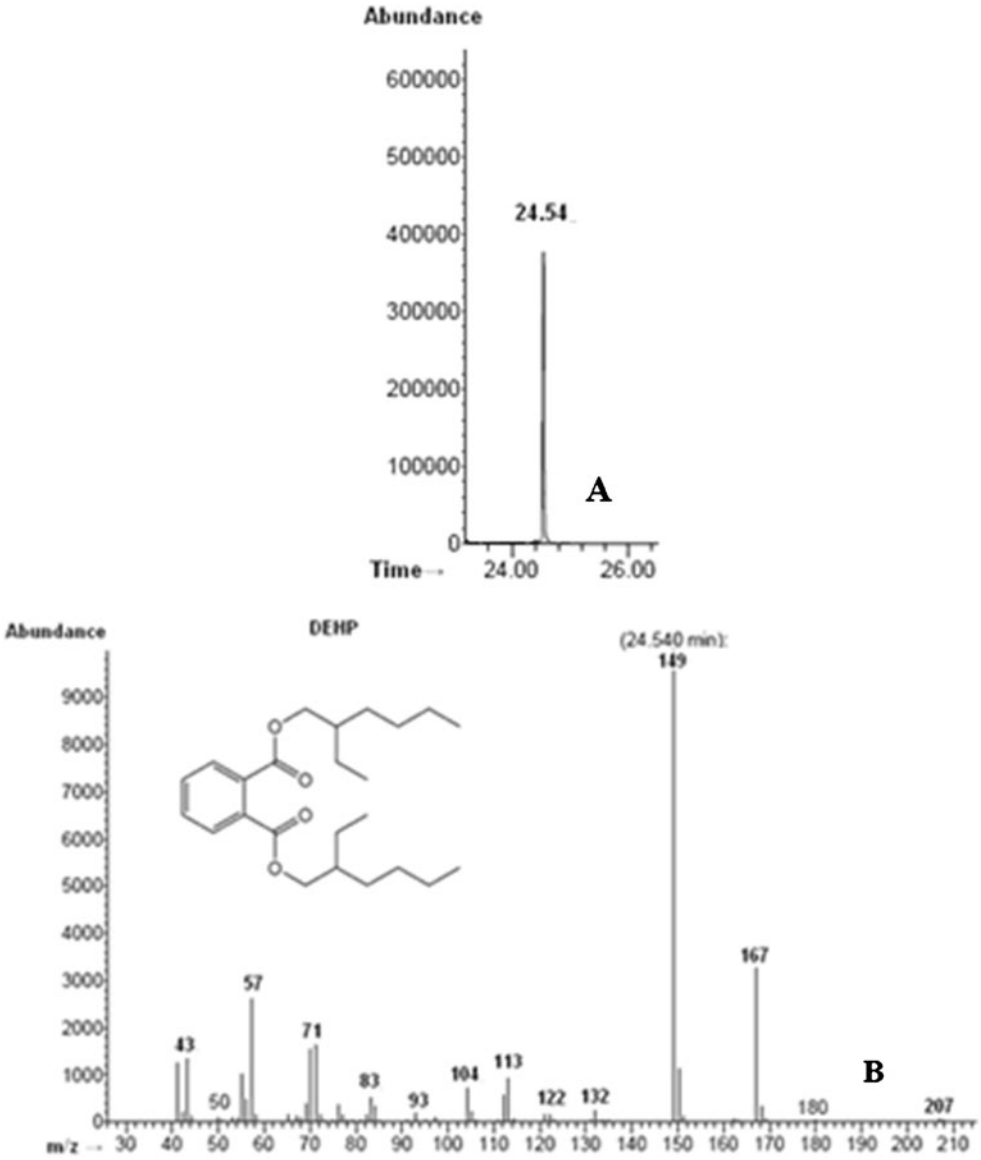


Fig. 3 GC-MS profile of DEHP, extracted in *n*-hexane from the PVC-BB, MBB: **a** GC profile and **b**. MS profile. **b** is the fragmentation profile of the corresponding GC peak at RT 24.54 min. The structure of DEHP is given at the inset

seen on the surface. The complete removal of DEHP makes the BB into a micro-porous network.

AFM pattern of BB

Figure 10 represents the AFM images of the outer surface of the BB (HPB) before (A) and after (B) DEHP extractions. In A, compact matrix is seen with less ridges and furrows. However in B, deep furrows indicating areas of DEHP removal are seen with much lesser sheet thickness.

DEHP extraction at boiling

Over 95 % of DEHP in BB was migrated into the solvent (data not shown) in the first h, when BB pieces were put in boiling *n*-hexane. Upon distillation entire DEHP was concentrated out from *n*-hexane, thus both components were separated. Taken together, all the three components in the extraction system, viz., microporous DEHP-free PVC sheet, DEHP and solvent were separated individually.

Discussion

As a part of our investigation to develop a strategy for the bioremediations of PAEs in plastics, we first explored the type of PAE present in three commercial BB available in India; followed by the detailed study on the estimation of PAE weight percentage, solvent-dependent migration profile of DEHP from BB, effect of temperature on its migration in to the surrounding solvent, determination of the diffusion coefficient and analyses of the physical texture of the BB before and after DEHP extraction. Owing to its simplicity and easiness for small sample handling, we opted

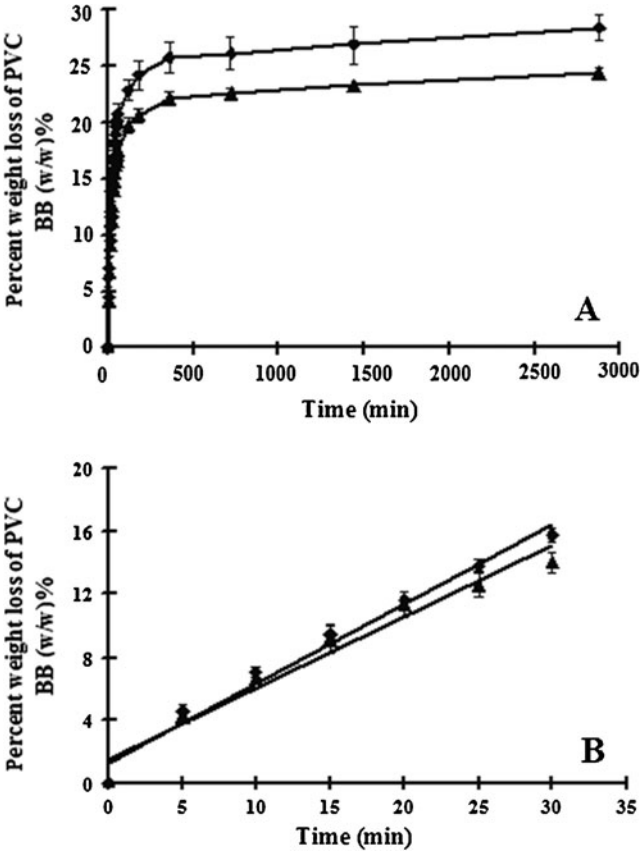


Fig. 4 Migration behavior of DEHP from two different PVC BB in to *n*-hexane at 40 °C: **a** for 48 h, (♦) HPB; (▲) MBB, and **b** initial 30 min showing linearity, (♦) HPB; (▲) MBB

for the conventional solvent extraction system in this study over other techniques like supercritical solvent extraction [18] or microwave-assisted extraction [19] techniques, which are mainly intended for large scale extraction. Initial studies showed that though all the BB tested contained about 34 % (w/w) PAE, DEHP was the sole PAE in two BBs, and the other contained a mixture of PAE. Hence, we used the BB contained only DEHP for further studies.

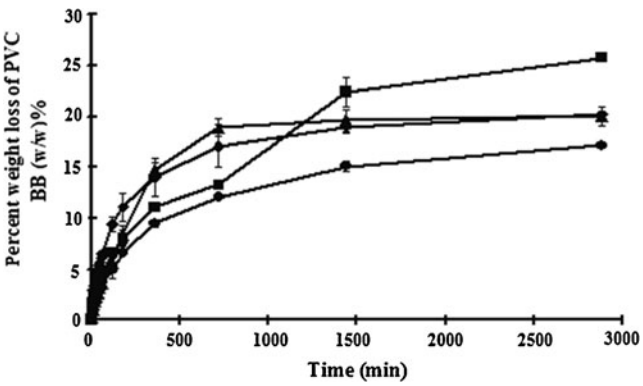


Fig. 5 Migration behavior of DEHP from the PVC-BB, MBB in to: (■) *t*-butyl alcohol; (●) ethanol; (▲) isopropyl alcohol; (♦) methanol at 40 °C

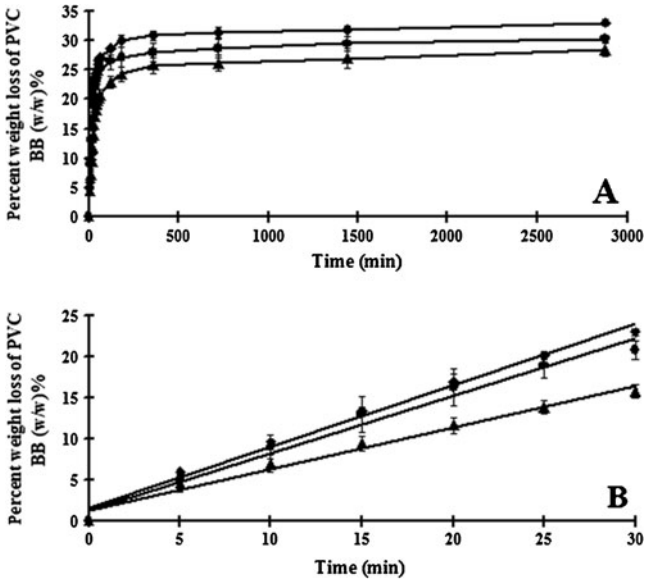


Fig. 6 Migration behavior of DEHP from the PVC-BB, HPB into *n*-hexane: **a** at varying temperatures, (▲) 40 °C; (●) 50 °C; (♦) 60 °C, and **b** initial 30 min showing linearity, (▲) 40 °C; (●) 50 °C; (♦) 60 °C

The presence of DEHP in BB was confirmed by UV spectrometry, TLC, GC and GC-MS. Though several PAE are used in the manufacture of PVC thermoplastics, DEHP is the predominant plasticizer being used for the manufacture of PVC-borne medical devices [9]. However, exact proportion of PAE used in each commercial PVC product is unclear. Ruzickova et al. [21] performed a detailed study in various European countries on 48 medical devices—

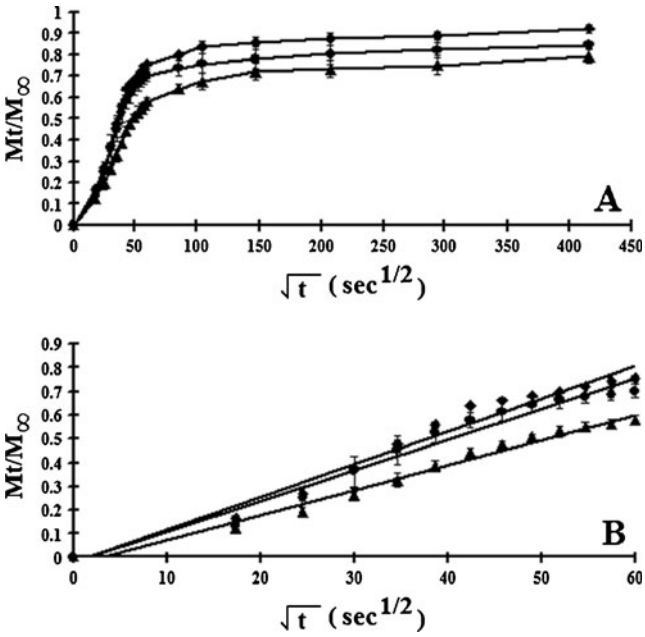


Fig. 7 Graph showing the linear relation between Mt/M_∞ vs. √t for the PVC-BB, HPB in *n*-hexane: **a** at varying temperatures for 48 h, (▲) 40 °C; (●) 50 °C; (♦) 60 °C, and **b** the initial 1h showing linearity, (▲) 40 °C; (●) 50 °C; (♦) 60 °C

Table 1 Diffusion coefficient of DEHP in *n*-hexane, estimated for two BB

BB	Thickness (cm)	Diffusion Coefficient, D (cm^2sec^{-1}) in *n*-hexane		
		Temperature (°C)		
		40	50	60
HPB	0.038	1.26×10^{-7}	1.92×10^{-7}	2.18×10^{-7}
MBB	0.049	1.61×10^{-7}	2.65×10^{-7}	3.14×10^{-7}

ranging from BB, injection sets, infusion and nutrition sets to tubing and masks—made from PVC plastic and found that DEHP used in such devices ranged between 17 and 41 % by weight as the plasticizer. We showed that the DEHP content in three brands of BB varied slightly from 33 to 35 % by weight. It further reflected that the release of DEHP into the surrounding solvent was directly proportional to the quantity of DEHP contained in the BB. Kim et al. [4] used thin sheets of PVC:DEHP blends in the ratio 30:70 and 40:60, respectively to show that the DEHP leaching in to the surrounding solvent was proportional to the amount of DEHP present in the PVC blend and temperature.

Our results show that surrounding solvent and temperature have crucial role in controlling the behavior of DEHP migration from the BB. At 40 °C, the intensity of DEHP

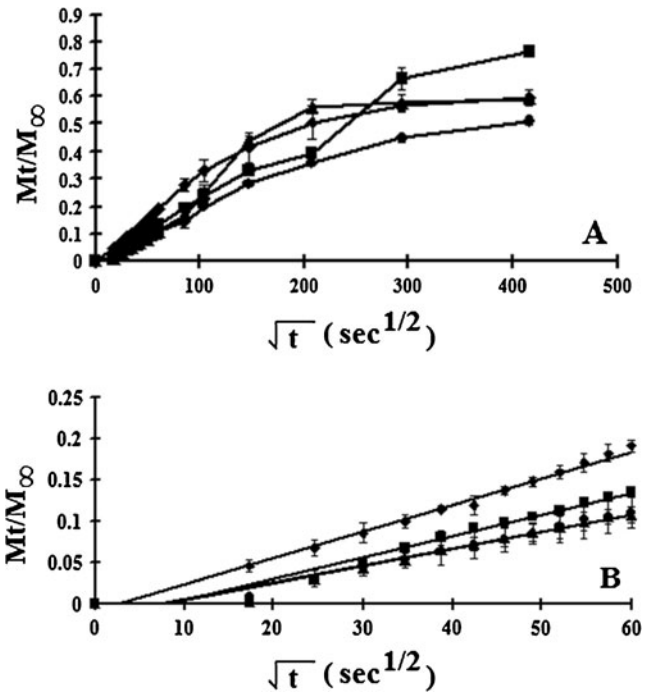


Fig. 8 Graph showing the linear relation between Mt/M_∞ vs. √t for the PVC-BB, MBB in four alcohols: **a** for 48 h at 40 °C, (■) *t*-butyl alcohol; (●) ethanol; (▲) isopropyl alcohol and (♦) methanol, and **b**. for initial 1 h showing the linearity, (■) *t*-butyl alcohol; (●) ethanol; (▲) isopropyl alcohol and (♦) methanol

Table 2 Diffusion coefficients of DEHP in different alcohols, estimated for two BB

BB	Thickness (cm)	Diffusion coefficient, D (cm^2sec^{-1}) in different alcohols at 40 °C			
		Methanol	*t*-butyl alcohol	Isopropyl alcohol	Ethanol
HPB	0.038	1.52×10^{-8}	5.43×10^{-9}	5.06×10^{-9}	6.18×10^{-9}
MBB	0.049	1.95×10^{-8}	1.24×10^{-8}	7.86×10^{-9}	7.99×10^{-9}

migration was in the order of *n*-hexane>methanol>*t*-butyl alcohol>isopropyl alcohol>ethanol. At this temperature, about 60 % DEHP was released into *n*-hexane during the initial hrs itself, while at 60 °C it increased up to 80 % in the initial 1 h of contact. Duvis et al. [22] showed that effect of UV irradiation and temperature on PAE migration from PVC in to various surrounding petroleum oils was greatly dependent on the liquid environment, i.e., higher viscous

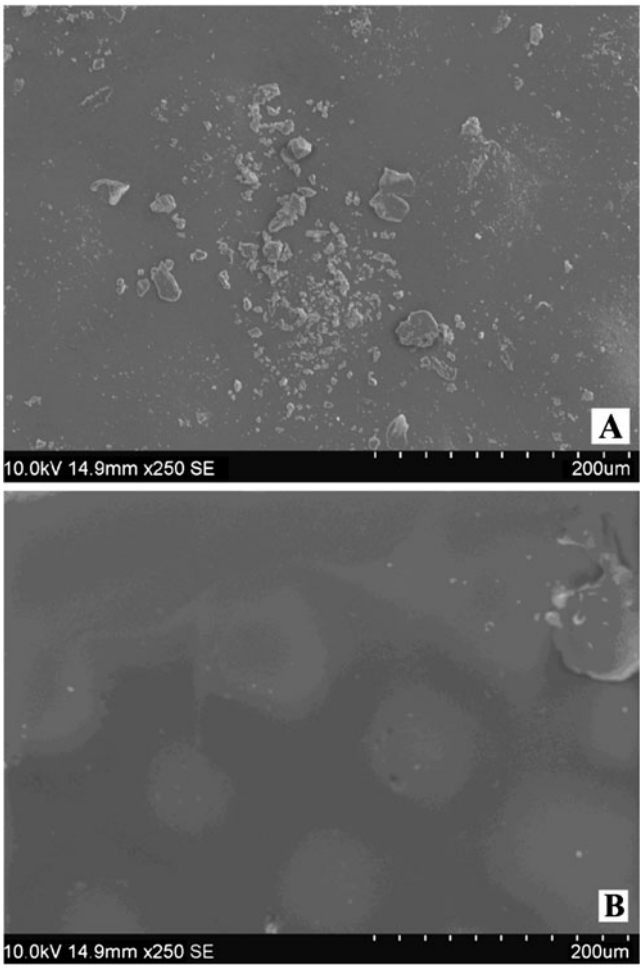


Fig. 9 **a** & **b**. SEM image of outer surface of the PVC-BB, HPB: **a** before extraction, it was soft and flexible with DEHP tightly packed in to the PVC net, and **b** after extraction of DEHP, the PVC became hard and stiff with micropores

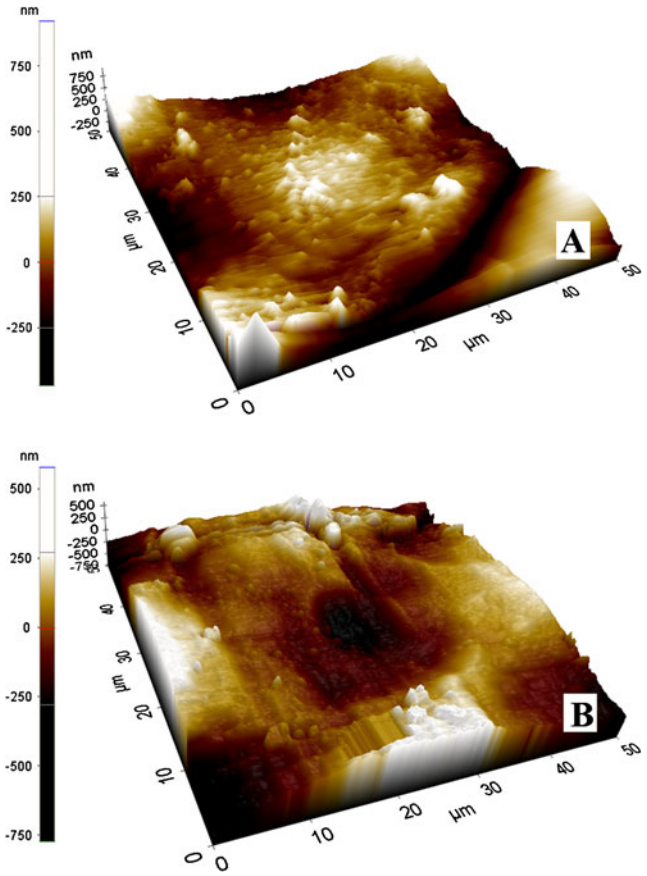


Fig. 10 **a** & **b** AFM images of the outer surface of the PVC-BB, HPB: **a** before extraction compactness with feeble ridges and furrows, and **b** after extraction ridges and furrows are very much prominent indicating the vacant areas left by the removal of DEHP. Operation details: head mode, NC-AFM; source, topography; data width, 256 (pxl); data height, 256 (pxl); X scan size, 50 (μm); Y scan size, 50 (μm); scan rate, 0.8 (Hz); Z servo grain, 5.05; set point, 0.2 (μm); amplitude, 0.68 (μm) and sel. frequency, 318.07E3 (Hz)

environment resisted migration than of lower viscosity. It further indicates that surface tension and dipole moment with coupled action of temperature might influence the penetration of the solvent into the BB thereby releasing DEHP. Migration behaviors of DEHP from plasticized PVC materials into different surrounding liquids have been reported [4, 23, 24]. Investigators made PVC sheets of their choice containing blends of PVC with different PAEs to study the migration behaviors in various solvent systems [4, 23, 24]. Using GC, Kim et al. [4] investigated the quantity, process and kinetics of DEHP migration in thin and flexible sheets of PVC:DEHP blends in water:ethanol mixture or acetonitrile, of which acetonitrile was more intense in inducing DEHP migration than water:ethanol mixture. Our data further depicts that after initial exponential rise in DEHP release, the migration profile was leveled off. It indicates that there existed a counter diffusion thereby attaining equilibrium. Takehisa et al. [25] observed similar phenomenon while studying with DEHP plasticized PVC medical tubing in a system where Tween 80 in various aqueous solutions co-existed at changing temperature and circulation speed. Haishima et al. [26] showed in another in vitro system using medical grade PVC tubing that the release of DEHP in to blood was reached up to 7.3 mg by 4 h circulation upon hemodialysis and pump-oxygenation therapy.

We also investigated the diffusion coefficient D, using Fickian approximation. Our data depict that D was directly proportional to the temperature of the surrounding medium, which in fact, attained equilibrium in about first 1 h itself. It suggests that upon rise in temperature, the PVC matrix gets loosened, facilitating faster liberation of the physically bound DEHP from the PVC network. In comparison, DEHP showed higher solubility in *n*-hexane than other solvents tested, which indicates faster migration in to *n*-hexane and thereby increased D values. This kinetics was in good conformity to the short time Fickian approximation, as proposed by Duvis et al. [22].

For physical confirmation of DEHP migration, we resorted to SEM and AFM techniques. We adopted noncontact mode of AFM analysis, because contact mode operation of the AFM probe induces surface erosion and false positive results [27]. Removal of DEHP bound to BB made the PVC a blank micro-porous sheet. The high resolution SEM and AFM microtopographs physically reveal that vacant spaces created after DEHP removal, leaving micro-spaces on blank PVC mat.

The novelty of this study is that we used commercial BB as a model to see its migration behavior in various solvents, thus to screen the presence of PAE in medical devices easily. Most of the DEHP contained in the BB migrated into the solvent in the first h upon boiling in the reflux condenser (data not shown), indicating *n*-hexane could be used as a preferred solvent for the fast screening of the hazardous DEHP in medical or other devices. Briefly, this study shows that *n*-hexane is the preferred solvent to extract DEHP or PAE from used or waste medical and allied products to obtain DEHP-free micro-porous PVC sheet, coupled with the recovery of pure DEHP and *n*-hexane by distillation inexpensively and eco-friendly, all for reuse.

Conclusions

Owing to its cost-effectiveness and easiness in small sample handling, we have used the conventional solvent extraction system in this study. PVC scrapes, bags, etc., containing PAE or specifically DEHP are usually dumped in to the soil or burnt, causing serious environmental and health concerns. During this unhealthy process, the DEHP will be leached in to the environment, which ultimately reaches into

the human system. As we described in the present study, the hazardous DEHP in used or waste medical and allied products can easily be extracted in to a solvent, preferably *n*-hexane, thereby making PVC component free. Upon distillation, the DEHP was concentrated and freed from the solvent used for extraction. The hazardous DEHP extracted from PVC could be either reused or biodegraded. Thus, all the three components in the DEHP-PVC-extraction system could be reused without much material loss, all afresh with a very little or no harm to the environment and human health.

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Conflict of interest The authors declare that there exists no conflict of interest.

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